

Enter information for sections 1, 2 and 3 - incorporate in the drawing set.

Must be typewritten

1 Location Information				
House No(s)	79 Street Name WINTHROP STREET			
Borough	BROOKLYN	Block	5045	Lot 72 BIN 3115681 CB No. 309
Job Type (NB, Alt, etc.)	New Building			

2a Applicant Information			
Last Name	FAYBYSHENKO	First Name	GENNADIY
Middle Initial	B.		
Business Name	CHOSEN MEP, CORP.		Business Email
Business Address	31-00 47TH AVENUE SUITE #3100		Business Telephone 347-766-5010
City	LONG ISLAND CITY	State	NY
Zip	11101		
Email	INFO@CHOSENMEP.COM		License Number 091045

2b Modeling Firm			
Last Name (lead modeler)	BEREGOVYKH		First Name (lead modeler)
Middle Initial	NIKITA		
Business Name	LIBER CREATIVE LLC		Email (lead modeler)
Business Address	2306 OCEAN AVENUE		Telephone (lead modeler)
City	BROOKLYN	State	NY
Zip	11229		

3a Energy Modeling Information	
Modeling Protocol (Sect 11 or Apx G)	Appendix G PRM
Modeling Software & version	EnergyPlus Version 25.1.0-5443.3.000
Weather File	NEW YORK LAGUARDIA
Proposed Unmet Load Hours	70.66
Baseline Unmet Load Hours	34.66
Gross Square Feet	12381.63
Modeled Square Feet	11690.791
Conditioned Square Feet	11169.925
Semi-heated Square Feet	0
Unconditioned Square Feet	520.866

3c List all Job Numbers Associated with this Model	
ARCHITECTURAL	B01304513-P1
MECHANICAL	B01304513-P6
PLUMBING	B01304513-P5

3b Energy Modeling Information per Modeled Square Feet			
Proposed	Total Energy Cost (\$/sf)	\$	2.19
	Site EUI (kBtu/sf)		29.83
	Source EUI (kBtu/sf)		76.08
	GHG (tCO2e/sf)		0.002527
Baseline	Total Energy Cost (\$/sf)	\$	3.46
	Site EUI (kBtu/sf)		80.63
	Source EUI (kBtu/sf)		143.03
	GHG (tCO2e/sf)		0.005511
% Savings	Total Energy Cost (\$/sf)		36.75%
	Site EUI (kBtu/sf)		63.00%
	Source EUI (kBtu/sf)		46.81%
	GHG (tCO2e/sf)		54.15%

Falsification of any statement is a misdemeanor under the NYC Administrative Building Code and is punishable by a fine or imprisonment, or both. It is unlawful to give to a city employee, or for a city employee to accept, any benefit, monetary or otherwise, either as a gratuity for properly performing the job or in exchange for special consideration. Violation is punishable by imprisonment or fine or both.

Name (please print)

Signature

Date

P.E. / R.A. Seal (apply seal, then sign and date over seal)

FOR INFORMATIONAL PURPOSES ONLY: NOT REQUIRED FOR ENERGY CODE COMPLIANCE

3d	Carbon Emissions Limit (LL97)			
	Occupancy	Total Gross SQFT per occupancy	tCO2e limit (2024)	tCO2e limit (2030)
	TOTAL			

3e	Carbon Emissions Information (LL97)		
		tCO2e Limits for 2024-2029	tCO2e Limits for 2030-2034
	tCO2e Limit (LL97)	0	0
	tCO2e for this Building	30	
	Does this building comply with LL97?	This building is subject to meeting LL97 of 2019. By 2024, this proposed design may not comply with the requirements of LL97 based on the estimated carbon emissions.	
	**Total carbon emissions (tCO2e) for this building is an estimate for informational purposes based on the proposed occupancy, gross square footage, and modeled annual energy usage. The performance of the building will be affected by construction quality, occupancy, and operational impacts. Actual carbon emissions may be higher or lower than this estimate. **		

4	Energy Modeling Protocol (select)
	Appendix G PRM

4b	Purchased Energy Rates & Compliance Documentation						
Fuel	Utility Rate Provider/Rate Structure (i.e. ConEd)	Virtual Utility Rate (\$/unit)	Baseline Design Total Charge (\$)	Virtual Utility Rate (\$/unit)	Proposed Design Total Charge (\$)	Supporting Doc. Location	Model Output Report
Electric	Con Edison	\$0.250	\$ 33,332.00	\$0.250	\$ 25,556.00	Energy Model (Appendix G)	BEPU, BEPU-P
Gas/Steam	Con Edison	\$1.450	\$ 7,070.00			Energy Model (Appendix G)	BEPU, BEPU-P
Other - (\$/kBTU)						Energy Model (Appendix G)	BEPU, BEPU-P
						COST SAVINGS	COMPLIES?
TOTAL			\$ 40,402.00		\$ 25,556.00	\$ 14,846.00	COMPLIES

4c.	Performance Cost Index - Appendix G ONLY	
	Baseline Building	Proposed Building
Building Performance Factor	0.670	Multifamily
Baseline Regulated Cost	\$ 39,679.75	\$ 24,833.75
Baseline Unregulated Cost	\$ 722.25	\$ 722.25
Total Energy Cost	\$ 40,402.00	\$ 25,556.00
Performance Cost Index	0.676	0.633

4d.	Performance Source Energy Index - Appendix G ONLY	
	Baseline Building	Proposed Building
Building Performance Factor	0.680	Multifamily
Baseline Regulated Source Energy	1,646,956	864,283
Baseline Unregulated Source Energy	25,136	25,136
Total Source Energy	1,672,092	889,419
Performance Source Energy Index	0.685	0.532

4e	Weighted BPF	
Space Classification	Modeled Area	Weighted BPF (energy cost)
All Other		Input modeled areas for each
Dormitory		Weighted BPF (source energy)
Healthcare/hospital		Input modeled areas for each
Hotel/motel		
Multifamily	11,691	
Office		
Restaurant		
Retail		
School		
Warehouse		
TOTAL	11,691	

4f.	Weighted Envelope Backstop (results from ComCheck)	
Allowable Margin (baseline)		
Envelope Design (proposed)		

INSTRUCTIONS: If the modeling parameters call for averaging 4 rotations, enter the data into each of the white columns. If the modeling protocol does not require averaging the 4 rotations, enter the data into only the column labeled "0'" (do not enter '0' into the columns labeled 90, 180, 270).

Baseline Rotations																
Baseline Model																
	0'			90'			180'			270'			Average of 4 rotations			Model Output Location (Report)
Annual usage (kwh/therm/kBTU)																
	kwh	therm	kBTU	kwh	therm	kBTU	kwh	therm	kBTU	kwh	therm	kBTU	kwh	therm	kBTU	
Interior Lighting	29,869	-	-										29,869	-	-	BEPU
Misc. Equip. Unregulated	2,889	-	-										2,889	-	-	BEPU
Misc. Equip. Regulated	-	-	-										-	-	-	BEPU
Space Heat	-	3,493	-										-	3,493	-	BEPU
Space Cool	20,828	-	-										20,828	-	-	BEPU
Heat Rejection	-	-	-										-	-	-	BEPU
Pumps & Misc	5,433	-	-										5,433	-	-	BEPU
Vent Fans	73,953	-	-										73,953	-	-	BEPU
Dom. Hot Water	75	1,384	-										75	1,384	-	BEPU
Exterior Lighting	278	-	-										278	-	-	BEPU
Exterior Misc.	-	-	-										-	-	-	BEPU
TOTAL	133,325	4,877											133,325	4,877		BEPU
Annual cost (\$)																
Electric	\$33,332.00												\$33,332.00			BEPU
Gas/Steam	\$7,070.00												\$7,070.00			BEPU
Other																BEPU
TOTAL	\$40,402.00												\$40,402.00			BEPU

Model Output Location (Report)	Category	Baseline Reports File	Proposed Reports File
BEPU / BEPU-P	Annual	Pages 2, 9, 20, 22, 102, 221	Pages 2, 10, 21, 23, 89, 205
SV-A / SV-A-P	HVAC	Pages 25, 203, 205-206, 208	Pages 26, 185, 188-191
ERV / ERV-W	Envelope	Pages 106	Pages 93
LV-C / LV-D	Lighting	Pages 193	Pages 176

5b	Energy Modeling Usage Summary											
Compliance based on Appendix G PRM												
	Baseline Model				Proposed Model							
	Electric Usage (kwh)	Gas/Steam Usage (Therm)	Other (i.e. chilled water) (kBtu)	% Usage	Electric Usage (kwh)	Gas/Steam Usage (Therm)	Other (i.e. chilled water) (kBtu)	% Usage	Energy Savings Per End Use (%)	Cost Savings per End Use (%)	Model Output Location (Report)	
Interior Lighting	29,869	-	-	11%	3,408	-	-	3%	15%	85%	BEPU, BEPU-P	
Misc. Equip. Unregulated	2,889	-	-	1%	2,889	-	-	3%	0%	0%	BEPU, BEPU-P	
Misc. Equip. Regulated	-	-	-	0%	-	-	-	0%	0%	0%	BEPU, BEPU-P	
Space Heat	-	3,493	-	37%	25,217	-	-	25%	44%	-81%	BEPU, BEPU-P	
Space Cool	20,828	-	-	8%	6,767	-	-	7%	8%	45%	BEPU, BEPU-P	
Heat Rejection	-	-	-	0%	-	-	-	0%	0%	0%	BEPU, BEPU-P	
Pumps & Misc	5,433	-	-	2%	5,319	-	-	5%	0%	0%	BEPU, BEPU-P	
Vent Fans	73,953	-	-	27%	17,675	-	-	17%	32%	181%	BEPU, BEPU-P	
Dom. Hot Water	75	1,384	-	15%	40,669	-	-	40%	0%	-131%	BEPU, BEPU-P	
Exterior Lighting	278	-	-	0%	281	-	-	0%	0%	0%	BEPU, BEPU-P	
Exterior Misc.	-	-	-	0%	-	-	-	0%	0%	0%	BEPU, BEPU-P	
TOTAL	133,325	4,877	-	100%	102,225	-	-	100%	100%	100%	BEPU, BEPU-P	

5c	Energy Related Design Features		
List energy related features in the design that <u>exceed</u> the mandatory/prescriptive requirements under each category.		List energy related features in the design that <u>do not meet</u> the prescriptive requirements under each category.	
Opaque Envelope	High-performance insulated roof assembly exceeding minimum NYCECC thermal requirements. <u>Insulated exterior wall assemblies with continuous</u>	Opaque Envelope	
Fenestration & WWR	High-performance triple-pane low-E windows with U-factor and SHGC exceeding prescriptive limits. <u>Window-to-wall ratio maintained below maximum</u>	Fenestration & WWR	
Ventilation Systems	Central energy recovery ventilator (ERV) providing heat recovery in excess of prescriptive requirements.	Ventilation Systems	
Cooling Systems	High-efficiency air-source heat pump systems exceeding minimum efficiency requirements.	Cooling Systems	
Heating Systems	High-efficiency air-source heat pump heating with variable-speed operation.	Heating Systems	
Service Water Heating	All-electric service water heating system with demand-controlled hot water recirculation.	Service Water Heating	
Lighting Power	Interior lighting power densities below prescriptive maximums using high-efficiency LED fixtures. <u>Exterior lighting power below prescriptive allowances.</u>	Lighting Power	
Lighting Controls	Automatic lighting controls including occupancy sensors and time-based controls in accordance with NYCECC.	Lighting Controls	
Other	Fully electric building design reducing fossil fuel use and associated emissions.	Other	

6	Above-Grade Wall & Fenestration Areas								
Compliance based on Appendix G PRM									
		Baseline Case			Proposed Case			Supporting Doc. Location	Model Output Report
		Other							
		Orientation	Window + Wall Area (ft2)	Vertical Glazing Area		Window + Wall Area (ft2)	Vertical Glazing Area		
(ft2)	(%)			(ft2)	(%)				
North	1800.479		667.147	37.1	1800.479	1078.436	59.9	EN-006	ERV, ERV-W
East	3209.583		22.712	0.7	3209.583	48.653	1.5	EN-006	ERV, ERV-W
South	1821.899		784.151	43.0	1821.899	1053.464	57.8	EN-006	ERV, ERV-W
West	3466.517		35.951	1.0	3466.517	81.59	2.4	EN-006	ERV, ERV-W
Total	10298.478	1509.961	14.7	10298.478	2262.143	22.0	EN-006	ERV, ERV-W	
Roof & Skylight Area	Roof + Skylight Area (ft2)	Skylight Area		Roof + Skylight Area (ft2)	Skylight Area		Supporting Doc. Location	Model Output Report	
		(ft2)	(%)		(ft2)	(%)			
	Total	2126.303	0	0.0	2126.303	0	0.0	EN-007	SV-A, SV-A-P

6b Vertical Fenestration													
Compliance based on Appendix G PRM													
Model Input Parameter	Category (Res/Non-Res)	Item #	Baseline Case				Proposed Case					Supporting Doc. Location	Model Output Report
			Description (from ASHRAE)	U-factor	SHGC	VLT	Description (from design)	U-factor	SHGC	VLT	% Overall Wall/Roof Area		
Vertical	Residential	1	Metal framing (fixed), below 95ft	U-0.57	0.39	0.45	Triple Pane Low-E (A-400)	0.3	0.3	0.452	22%	A-400.00	ERV, ERV-W
Vertical		2											
Vertical		3											
Vertical		4											
Vertical		5											
Vertical		6											
Vertical		7											
Vertical		8											
Vertical		9											
Vertical		10											
Vertical		11											
Vertical		12											
Skylight		1											
Skylight		2											
Shading Devices		Shading projections, manual shading devices, or self-shading have been modeled.				List any permanent or auto-controlled shading devices:				Total % WWR	22%		
		Any shading by adjacent structures has been modeled identically to the proposed case.								Total % skylt			

6c	Opaque Elements								
Compliance based on Appendix G PRM									
Model Input Parameter	Space-Conditioning Category	Item #	Baseline Case		Proposed Case		% of above-grade wall	Supporting Doc Location	Model Output Report
			Description	U-factor/ C-factor/ F-factor	Description	Assembly U-factor/ C-factor/ F-factor			
Roof Construction	Residential	1	Insulation Entirely Above Deck	U-0.063	Roof assembly (TPO + 6" rigid foam + 2" GRC + 2" GRC)	0.024	100	A-500	ERV, ERV-W
		2	Solar Reflectance	SR =					
Above-Grade Exterior Wall Construction (Values on this tab should align with modeled inputs. For additional rows, use tab 6c2)	Residential	1	Steel-Framed	U-0.064	Ext wall (brick veneer + EIFS R-8 + spray foam)	0.047	78%	A-500	ERV, ERV-W
		2							
		3							
		4							
		5							
		6							
		7							
		8							
		9							
		10							
Below-Grade Exterior Wall Construction	Semiheated	1	Below-Grade Wall	C-1.140	Below-grade concrete	0.048	100	A-500	ERV, ERV-W
		2							
Exposed Floor Construction		1							
Slab-On-Grade Floors	Residential	1	Unheated	F-0.730	Slab on grade (unheated)	0.73	100	A-500	ERV, ERV-W
Opaque Doors	Residential	1	Swinging	U-0.700	Exterior doors (U=0.30, SHGC=0.30) per A-400	0.3	1	A-500	ERV, ERV-W
		2							
						Total % Opaque Wall	78%		

6c	Opaque Elements - Weighted Average								
Compliance based on Appendix G PRM									
Model Input Parameter	Space-Conditioning Category	Item #	Baseline Case		Proposed Case			Supporting Doc Location	Model Output Report
			Description	U-factor/ C-factor/ F-factor	Description	Assembly U-factor/ C-factor/ F-factor	% of above-grade wall		
Roof Construction		1							
		2							
		3							
		4							
		5							
		6							
		7							
Above-Grade Opaque Wall Construction		1							
		2							
		3							
		4							
		5							
		6							
		7							
		8							
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		28							
		29							
		30							
		31							
		32							
		33							
		34							
		35							
						Total % Opaque Roof			
						Total % Opaque Wall			

6e	Exterior Lighting Power			
Compliance based on Appendix G PRM				
	Baseline Design (Watts)	Proposed Design (Watts)	Supporting Doc. Location	Model Output Report
Tradable Lighting Power	64	64	RCP-001	LV-C, LV-D
Non-Tradable Lighting Power				
Base Site Allowance	0			
Total Lighting Power	64	64	RCP-001	LV-C, LV-D

3	M districts, except MX; C districts, except C5, C6 and C overlays on R
0	Base Site Allowance (watts)

Area Type	Area Parameters			Length or Area (FT or SQFT)	Proposed Watts	Baseline Watts	Unit Allowances
Tradable Surfaces	Uncovered Parking Areas	Total area of uncovered parking and drive areas (sq.ft):					0.15 W/SF
	Building Grounds	Length of walkways less than 10' wide					1.00 W/SF
		Area of walkways greater than 10' wide					0.20 W/SF
		Plaza areas (sqft)					0.20 W/SF
		Special feature areas (sqft)					0.20 W/SF
		Dining areas (sqft)					W/SF
		Area of stairways (sqft)					1.00 W/SF
		Area of pedestrian tunnels (sqft)					W/SF
		Area of landscaping (sq.ft):					W/SF
	Building Entrances & Exits	Total door width of pedestrian and vehicular entrances and exits (ft):					30.00 W/FT
		Area of entry canopies (sq.ft):					W/SF
		Loading Docks (sq.ft):					W/SF
	Sales canopies	Free standing and attached (sq.ft):					W/SF
	Outdoor sales areas	Open areas (including vehicle sales and lots) (sq.ft):					0.50 W/SF
		Total length of street frontage for vehicle sales (ft):					20.00 W/FT
Non Tradable Surfaces	Building façades	Area of façade #1 (sqft)		Length of façade #1 (ft)			W/FT
		Area of façade #2 (sqft)		Length of façade #2 (ft)			W/FT
		Area of façade #3 (sqft)		Length of façade #3 (ft)			W/FT
		Area of façade #4 (sqft)		Length of façade #4 (ft)			W/FT
	ATMs	# of Locations:		Total # of ATMs:			
	Guarded facility entrances	Area of uncovered entrances and gatehouse inspection areas at guarded facilities (sq.ft):					W/FT
	Emergency Vehicle Loading	Area of uncovered loading areas for emergency service vehicles (sq.ft):					W/FT
	Drive-throughs	# of drive-throughs:					Watts/drive thru
	Parking near 24-hour retail entrances (main entrances)	# of main entrances:					Watts/entrance

6g.	Service Hot Water Systems					
Compliance based on Appendix G PRM						
Model Input Parameter	Baseline Design	Units	Proposed Design	Units	Supporting Doc Location	Model Output Report
	Multifamily					
System Type & Fuel	Gas Storage Water Heater		Electric Storage Water Heater(s) + POU		PLUMBING SCHEDULES	SV-A, SV-A-P
Input Rating		kBtu/h	29.5	kW	PLUMBING SCHEDULES	SV-A, SV-A-P
Efficiency	0.8	EF	0.93	Et	PLUMBING SCHEDULES	SV-A, SV-A-P
Storage Volume		Gal	395	Gal	PLUMBING SCHEDULES	SV-A, SV-A-P
Storage Temperature	120	°F	120	°F	PLUMBING SCHEDULES	SV-A, SV-A-P
Peak HW Demand		GPM	0.58	GPM	PLUMBING SCHEDULES	SV-A, SV-A-P
Number of Primary DHW pumps		#	8	#	PLUMBING SCHEDULES	SV-A, SV-A-P
Primary DHW Pump Power		kW	4.18	kW	PLUMBING SCHEDULES	SV-A, SV-A-P
Primary DHW Pump Control	Constant (baseline default)		Demand controlled / continuous (per MEP)		PLUMBING SCHEDULES	SV-A, SV-A-P

6h HVAC and SHW Narrative	Please provide a narrative of the building's heating, cooling, ventilation & service hot water systems
The proposed building is a new multifamily residential building. Space heating and cooling are provided by individual air-source split-system heat pumps serving each dwelling unit, with outdoor condensing units located on the roof and indoor air handlers within the conditioned spaces. Outdoor ventilation air is provided by a central energy recovery ventilator (ERV) in accordance with ASHRAE 62.1, with exhaust from kitchens and bathrooms discharged to the roof. Service hot water is provided by electric storage water heaters with supplemental point-of-use electric water heaters. The system operates at a storage temperature of 120°F and includes demand-controlled hot water recirculation. For energy code compliance using ASHRAE 90.1-2019 Appendix G, the baseline building is modeled with a gas-fired storage water heater, while the proposed design reflects the all-electric service water heating system shown on the plumbing drawings.	

HVAC Cover Sheet					Proposed Building Classification (auto-populated + drop-down)			Baseline systems correspond to Appendix G PRM
Building Area Type (Only used for Appendix G)	Proposed System Design	System Name in Mechanical Schedule	System Name in Model Reports	System Name in Model Reports	Condenser Cooling Source	Heating System Classification	Zone Classification	
Residential	Packaged/split-system heat pump				Air/None	Heat Pump	Single-Zone Residential	System 1: PTAC

Baseline System	Minimum Efficiencies	Fan Power	Heating System Classification
System 1: PTAC	Cool: 3.2 COP -- Heat: Boiler	Supply CFM * .3	Hot-water fossil fuel boiler

System Count
1

Air-Side HVAC						
	HVAC System / Group (BASELINE DESIGN)		HVAC System / Group (PROPOSED DESIGN)			
	Description	Units	Description	Units	Supporting Doc. Location	Model Output Report
System Description						
System Name in Model	HOT WATER LOOP / PTAC		PACKAGED ROOFTOP HEAT PUMP		MECHANICAL EQUIPMENT SCHEDULES	SV-A, SV-A-P
# of Similar Systems	1		1		MECHANICAL EQUIPMENT SCHEDULES	SV-A, SV-A-P
Total Cooling Capacity	924846.659	BTU / h	281873.608	BTU/h	MECHANICAL EQUIPMENT SCHEDULES	SV-A, SV-A-P
*Table 6.8.1 Unitary Cooling Capacity Range		BTU / h				
*Unitary Cooling Eff.	8.89	EER	10.84	EER	MECHANICAL EQUIPMENT SCHEDULES	SV-A, SV-A-P
*Unitary Cooling Part-load Eff. (if applicable)	9.1	IEER	11	IEER	MECHANICAL EQUIPMENT SCHEDULES	SV-A, SV-A-P
Total Heating Capacity	617087.691	BTU/h	293439.54	BTU/h	MECHANICAL EQUIPMENT SCHEDULES	SV-A, SV-A-P
*Table 6.8.1 Unitary Heating Capacity Range		Et	Et			
*Unitary Heating Efficiency	NA- boiler dependent	NA	1.94	COP	MECHANICAL EQUIPMENT SCHEDULES	SV-A, SV-A-P
*Fan Control	Single- or two-speed fan		Variable-speed fan		MECHANICAL EQUIPMENT SCHEDULES	SV-A, SV-A-P
Supply Airflow	28075.16	cfm	9259.506	cfm	MECHANICAL EQUIPMENT SCHEDULES	SV-A, SV-A-P
Outdoor Airflow		cfm		cfm		
*Demand Control Ventilation	No		No			
Economizer provided?						
*Economizer High-Limit Shutoff (°F)						
Exhaust Air Energy Recovery System Provided?	No		Yes		MECHANICAL EQUIPMENT SCHEDULES	SV-A, SV-A-P
*Exhaust Air Energy Recovery Effectiveness			0.65		MECHANICAL EQUIPMENT SCHEDULES	SV-A, SV-A-P
Fan Brake Horsepower	11.32097553	hp	2.508918226	hp	MECHANICAL EQUIPMENT SCHEDULES	SV-A, SV-A-P
Fan Motor Efficiency	80	%	90	%	MECHANICAL EQUIPMENT SCHEDULES	SV-A, SV-A-P
Supply Fan Power		kW	1.8709	kW	MECHANICAL EQUIPMENT SCHEDULES	SV-A, SV-A-P
Return Fan Power		kW		kW		
Exhaust Fan Power		kW	0.15	kW	MECHANICAL EQUIPMENT SCHEDULES	SV-A, SV-A-P
Relief Fan Power		kW		kW		
Allowed/Total Fan Power:		kW		kW		
EIR	0.38		0.31			
HIR			0.52			
% OA						
Other (describe)						